
UNIT 11 CLUSTERING

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11.0 INTRODUCTION

In the earlier unit, we have studied the Classification in Data Mining. We have covered the introductory concepts and general approach to classification, applications of classification models, various classifiers and their underlying principles and model evaluation and selection aspects.

This unit covers another important concept known as Clustering. **Clustering** is the process by which we create groups in a data, like customers, products, employees, text documents, in such a way that objects falling into one group exhibit many similar properties with each other and are different from objects that fall in the other groups that got created during the process

It is a technique used to turn a collection of unrelated abstract objects into a set of related classes. It is a technique for dividing large collections of data or objects into smaller groups called clusters. As a stand-alone tool, it provides insight into data distribution and can be used as a pre-processing step for other algorithms or as a pre-processing step in its own right.

We will study overview of clustering, clustering methods, partitioning method, hierarchical clustering and outlier analysis.

11.1 OBJECTIVES

After going through this unit, you shall be able to:

- understand Clustering in Data Mining;

- list various types of Clustering Methods;
- understand and demonstrate various Clustering Methods;
- understand the k-Means algorithm and k-Medoids which falls under partitioning method;
- explain the Agglomerative and Divisive methods of hierarchical clustering, and
- discuss outlier analysis.

11.2 CLUSTERING – AN OVERVIEW

Clustering is the process of grouping a collection of objects (usually represented as points in a multidimensional space) into classes of similar objects. Cluster analysis is a very important tool in data analysis. It is a set of methodologies for automatic classification of a collection of patterns into clusters based on similarity. Intuitively, patterns within the same cluster are more similar to each other than patterns belonging to a different cluster. It is important to understand the difference between clustering (unsupervised classification) and supervised classification.

Typical pattern clustering activity involves the following steps:

- pattern representation (including feature extraction and/or selection),
- definition of a pattern proximity measure appropriate to the data domain,
- clustering
- data abstraction and
- assessment of output.

Cluster analysis is an exploratory discovery process. It can be used to discover structures in data without providing an explanation/interpretation.

Cluster analysis includes two major aspects: *clustering* and *cluster validation*. Clustering aims at partitioning objects into groups according to a certain criteria. To achieve different application purposes, a large number of clustering algorithms have been developed. While due to there are no general purpose clustering algorithms to fit all kinds of applications, thus, it is required an evaluation mechanism to assess the quality of clustering results that produced by different clustering algorithms or a clustering algorithm with different parameters, so that the user may find a fit cluster scheme for a specific application. The quality assessment process of clustering results is regarded as cluster validation. Cluster analysis is an iterative process of clustering and cluster verification by the user facilitated with clustering algorithms, cluster validation methods, visualization and domain knowledge to databases.

Clustering is an unsupervised-based algorithm that divides a set of data points into clusters so that the objects are all part of the same group. Using clustering, data can be subdivided into smaller, more manageable groups. Data from each of these subgroups are grouped together into a single cluster.

- To put it simply, a cluster is a group of related things.
- It's a grouping of objects where the distance between any two members is less than the distance between any two people.

- A multidimensional space segment with a high density of objects that is related to the other segments.

A simple example of Clustering is illustrated in Figure 1 as shown below:

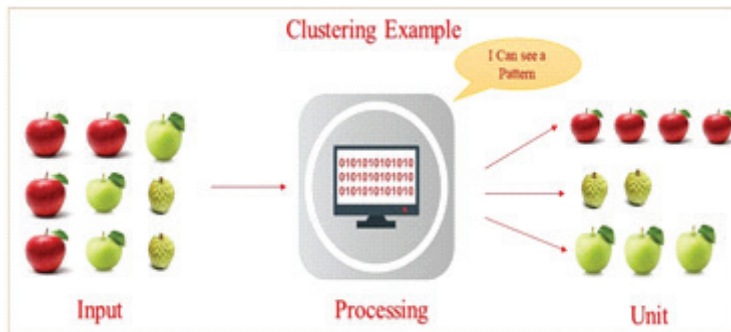


Figure 1: Example of Clustering

Clustering helps in organizing huge voluminous data into clusters and displays interior structure of statistical information. Clustering is the intent of segregating the data into clusters. Clustering improves the data readiness towards artificial intelligence techniques. Process for clustering, exhibits knowledge discovery in data, It is used either as a stand-alone tool to get penetration into data distribution or as a pre-processing step for other algorithm.

11.2.1 Applications of Cluster Analysis in Data Mining

Following are some of the applications of Cluster Analysis:

- Clustering analysis is widely utilized in a variety of fields, including data analysis, market research, pattern identification, and image processing.
- It aids data discovery by assigning documents on the Internet.
- Credit card fraud detection relies on clustering.
- Cluster analysis is a data mining function that provides insight into data distribution so that the properties of each cluster may be analyzed.
- It can be used in biology to figure out plant and animal taxonomies, classify genes with similar functions, and have a better understanding of population structure.
- Earth observation databases use this data to identify similar land regions and to group houses in a city based on house type, value, and geographic position.
- It is the backbone of search engine algorithms, where objects that are similar to each other must be presented together and dissimilar objects should be ignored. Also, it is required to fetch objects that are closely related to a search term, if not completely related.
- A similar application of text clustering like search engine can be seen in academics where clustering can help in the associative analysis of various documents – which can be in-turn used in – plagiarism, copyright infringement, patent analysis etc.
- Used in image segmentation in bioinformatics where clustering algorithms have proven their worth in detecting cancerous cells from various medical imagery – eliminating the prevalent human errors and other bias.

- OTT platforms are using clustering in implementing movie recommendations for its users.
- News summarization can be performed using Cluster analysis where articles can be divided into a group of related topics.
- Clustering is used in getting recommendations for sports training for athletes based on their goals and various body related metrics and assign the training regimen to the players accordingly.
- Marketing and sales applications use clustering to identify the Demand-Supply gap based on various past metrics – where a definitive meaning can be given to huge amounts of scattered data.
- Various job search portals use clustering to divide job posting requirements into organized groups which becomes easier for a job-seeker to apply and target for a suitable job.
- Resumes of job-seekers can be segmented into groups based on various factors like skill-sets, experience, strengths, type of projects, expertise etc., which makes potential employers connect with correct resources.
- Clustering effectively detects hidden patterns, rules, constraints, flow etc. based on various metrics of traffic density from GPS data and can be used for segmenting routes and suggesting users with best routes, location of essential services, search for objects on a map etc.
- Satellite imagery can be segmented to find suitable and arable lands for agriculture.
- Clustering can help in getting customer persona analysis based on various metrics of Recency, Frequency, and Monetary metrics and build an effective User Profile – in-turn this can be used for Customer Loyalty methods to curb customer churn.
- Document clustering is effectively being used in preventing the spread of fake news on Social Media.
- Website network traffic can be divided into various segments and heuristically when we can prioritize the requests and also helps in detecting and preventing malicious activities.
- Eateries are using clustering to perform Customer Segmentation which helped them to target their campaigns effectively and helped increase their customer engagement across various channels.

11.3 CLUSTERING METHODS

For a successful grouping there are two major goals – (i) similarity between one data point with another (ii) distinction of those similar data points with others which most certainly, heuristically differ from those points.

The basis of such divisions begins with our ability to scale large datasets and that's a major beginning point for us. This data contains different kinds of attributes like categorical data, continuous data etc.. Dealing with these is the second challenge. The next challenge is the multidimensional data. The clustering algorithm should successfully cross this hurdle as well.

The clusters not only there to distinguish data points but also they should be inclusive. Sure, a distance metric helps a lot but the cluster shape is often limited to being a geometric shape and many important data points get excluded. This problem too needs to be taken care of.

Also data is highly “noisy” in nature as many unwanted features have been residing in the data which makes it rather Herculean task to bring about any similarity between the data points, leading to the creation of improper groups. As we move towards the end of the line, we are faced with a challenge of business interpretation. The outputs from the clustering algorithm should be understandable and should fit the business criteria and address the business problem correctly.

To address the challenges such as scalability, attributes, dimensional, boundary shape, noise, and interpretation stated above there are various types of clustering methods to solve one or many of these problems.

The following are various types of Clustering methods:

- Partitioning Method
- Hierarchical Method
- Density-based Method
- Grid-Based Method
- Model-Based Method
- Constraint-based Method

11.3.1 Partitioning Method

Partitioning methods breaks down the data into a set of different clusters. Given n objects, this method produces k clusters of data where $k < n$ clusters of data and using an iterative relocation method. Algorithms used in partitioned clustering are k-means algorithm and k-medoids algorithm.

11.3.2 Hierarchical Method

Hierarchical methods, decomposes n number of data set into groups forming hierarchy of clusters. The tree like structural representation of hierarchical based clustering is called Dendrogram diagram. The root of dendrogram represents the entire dataset and leaves represent each individual cluster present in the dataset. The clustering results are obtained by taking dendrogram at different levels. The two approaches of hierarchical based clustering, Agglomerative (Bottom-up) method and Divisive (Top-down) method.

11.3.3 Density-Based Method

Density based clustering is clustering of database based on the densities (i.e. local cluster criterion). It has major features such as:

- It discovers the arbitrary shape cluster.
- It handles the noise data.
- It examines only the local region to justify the density.
- To termination the process it requires the density parameters.

It is categorized into two types namely:

- (a) **Density Based Connectivity:** This includes clustering technique such as DBSCAN (Density based Spatial Clustering of Applications with Noise) and DBCLASD.
- (b) **Density based Function:** DENCLUE (Density Based Clustering) method density clusters are obtained based on some functions.

11.3.4 Grid Based Method

The idea of grid-based clustering methods is based on the clustering oriented query answering in multilevel grid structures. The upper level stores the summary of the information of its next level, thus the grids make cells between the connected levels. Many grid-based methods have been proposed, such as STING (Statistical Information Grid Approach), CLIQUE, and the combination of grid-density based technique. The grid-based methods are efficient on clustering data with the complexity of $O(N)$. However the primary issue of grid-based techniques is how to decide the size of grids. This depends on the user's experience.

11.3.5 Model Based Clustering Method

Model-based clustering methods are based on the assumption that data are generated by a mixture of underlying probability distributions, and they optimize the fit between the data and some mathematical model, for example statistical approach, neural network approach and other AI approaches. When facing an unknown data distribution, choosing a suitable one from the model based candidates is still a major challenge. On the other hand, clustering based on probability suffers from high computational cost, especially when the scale of data is very large.

11.3.6 Constraint Based Method

The clustering process, in general, is based on the approach that the data can be divided into an optimal number of “unknown” groups. The underlying stage of all the clustering algorithms is to find those hidden patterns and similarities, without any intervention or predefined conditions. However, in certain business scenarios, we might be required to partition the data based on certain constraints. Here is where a supervised version of clustering machine learning techniques comes into play.

A constraint is defined as the desired properties of the clustering results, or a user's expectation on the clusters so formed – this can be in terms of a fixed number of clusters, or, the cluster size, or, important dimensions (variables) that are required for the clustering process. Usually, tree-based algorithms like Decision Trees, Random Forest, and Gradient Boosting, etc. are made use of to attain constraint-based clustering. A tree is constructed by splitting without the interference of the constraints or clustering labels. Then, the leaf nodes of the tree are combined together to form the clusters while incorporating the constraints and using suitable algorithms.

In the next section, let us study the algorithms available in partitioning method.

11.4 PARTITIONING METHOD

This method is one of the most popular choices for analysts to create clusters. This is also known as Supervised Clustering method. In partitioning clustering, the clusters are partitioned based upon the characteristics of the data points. We need to specify

the number of clusters to be created for this clustering method. These clustering algorithms follow an iterative process to reassign the data points between clusters based upon the distance. The algorithms that fall into this category are as follows:

11.4.1 k-Means Algorithm

This algorithm can be used to cluster the data set thus forming clusters repeatedly. It is one of the unsupervised and iterative algorithms. The main aim of this algorithm is to find out the location of the clusters thus minimizing the distance between the cluster and the data set. This algorithm also called “Lloyd’s algorithm” where m data set are clustered to form some number of clusters say k , where each of the data set belongs to the closer mean cluster.

Algorithm

1. Define the number of clusters (k) to be produced and identical data point centroids.
2. The distance from every data point to all the centroids is calculated and the point is assigned to the cluster with a minimum distance.
3. Follow the above step for all the data points.
4. The average of the data points present in a cluster is calculated and can set new centroid for that cluster.
5. Until desired clusters are formed repeat Step 2.

The initial centroid is selected randomly and thus the resulting clusters have larger influence on them. Complexity of k-means algorithm is $O(tkn)$ where n - total data set, k -clusters formed, t -iterations in order to form cluster.

Advantages

- Effortless implementation process.
- Dense clusters are produced when clusters are spherical when compared to hierarchical method.
- Appropriate for large databases.

Disadvantages

- Inappropriate for clusters with different density and size.
- Equivalent results are not produced on iterative run.
- Euclidean distance measures can weigh unequally due to underlying factors.
- Unsuccessful for non-linear data set and categorical data.
- Noisy data and outliers are difficult to handle.

11.4.2 k-Medoids or PAM (Partitioning Around Medoids)

It is similar in process to the K-means clustering algorithm with the difference being in the assignment of the center of the cluster. In PAM, the medoid of the cluster has to be an input data point while this is not true for K-means clustering as the average of all the data points in a cluster may not belong to an input data point.

In this algorithm, each of the cluster’s is described by one of the objects which is located near the centroid of the cluster. The repeated process of changing described

objects by non-described objects continues until the resulting cluster is improved. The value is predicted using cost function which measures the variance between an object and described object of a cluster.

The algorithm is implemented in two steps:

Build: Initial medoids are innermost objects.

Swap: A function can be swapped by another function until the function can no longer be reduced.

Algorithm

1. Initially choose m random points as initial medoids from given data set.
2. For every data point assign a closest medoid by distance metrics.
3. Swapping cost is calculated for every chosen and unchosen object given as TC_{ns} where s is selected and n is non-selected object.
4. If $TC_{ns} < 0$, s is replaced by n
5. Until there is no change in medoids, repeat 2 and 3.

Four characteristics to be considered are:

- Shift-out membership: Movement of an object from current cluster to another is allowed.
- Shift-in membership: Movement of an object from outside to current cluster is allowed.
- Update the current medoids: Current medoid can be replaced by a new medoid.
- No change: Objects are at their appropriate distances from cluster.

Advantages

- Effortless understanding and implementation process.
- Can run quickly and converge in few steps.
- Dissimilarities between the objects is allowed.
- Less sensitive to outliers when compared to k-means.

Disadvantages

- Initial sets of medoids can produce different clustering's. It is thus advisable to run the procedure several times with different initial sets.
- Resulting clusters may depend upon units of measurement. Variables of different magnitude can be standardized.

In the following section, let us focus on the algorithms available in the hierarchical method.

11.5 HIERARCHICAL METHOD

This method decomposes a set of data items into a hierarchy. Depending on how the hierarchical breakdown is generated, we can put hierarchical approaches into different categories. Following are the two approaches;

- Agglomerative Approach
- Divisive Approach

11.5.1 Agglomerative Approach

This Algorithm is also referred as Bottom-up approach. This approach treats each and every data point as a single cluster and then merges each cluster by considering the similarity (distance) in each individual cluster until a single large cluster is obtained or when some condition is satisfied.

Algorithm

1. Initialize all n data points into N individual clusters.
2. Find the cluster pairs with the least distance (closest distance) and combine them as one single cluster.
3. Calculate pair-wise distance between the clusters at present that is the new formed cluster and the priority available clusters.
4. Repeat steps 2 and 3 until all data samples are merged into a single large cluster of size N

Advantages

- Easy to identify nested clusters.
- Gives better results and ease in implementation.
- They are suitable for automation.
- Reduces the effect of initial values of cluster on the clustering results.
- Reduces the computing time and space complexity.

Disadvantages

- It can never undo what was done previously.
- Difficulty in handling different sized clusters and convex shapes lead to increase in time complexity
- There is no direct minimization of objective function.
- Sometimes there is difficulty in identifying the exact number of clusters by the Dendrogram.

11.5.2 Divisive Approach

This approach is also referred as the top-down approach. In this, we consider the entire data sample set as one cluster and continuously splitting the cluster into smaller clusters iteratively. It is done until each object in one cluster or the termination condition holds. This method is rigid, because once a merging or splitting is done, it can never be undone.

Algorithm

1. Initially, initiate the process with one cluster containing all the samples.
2. Select a largest cluster from the cluster that contains widest diameter.
3. Detect the data point in the cluster found in step 2 with the minimum average

similarity to the other elements in that cluster.

4. The first element to be added to the fragment group is the data samples found in step 3.
5. Detect the element in the original group which has the highest average similarity with the fragment group.
6. If the average similarity of element obtained in step 5 with the fragment group is greater than its average similarity with the original group then assign the data sample to the fragment group and go to step 5; otherwise do nothing;
7. Repeat the step 2 to 6 until each data point is separated into individual clusters

Advantage

- It produces more accurate hierarchies than bottom-up algorithm in some circumstances.

Disadvantages

- Top down approach is computationally more complex than bottom up approach because we need a second flat clustering algorithm.
- Use of different distance metrics for measuring distance between clusters may generate different results.

Let us study DBSCAN algorithm pertaining to the Density-based method in the next section.

11.6 DENSITY BASED METHOD

The primary idea of density-based methods is that for each point of a cluster the neighborhood of a given unit distance contains at least a minimum number of points, i.e. the density in the neighborhood should reach some threshold. However, this idea is based on the assumption of that the clusters are in the spherical or regular shapes.

11.6.1 DBSCAN(Density-Based Spatial Clustering of Applications with Noise)

DBSCAN was proposed to adopt density-reachability and density connectivity for handling the arbitrarily shaped clusters and noise. In DBSCAN, a cluster is defined as group of data that is of highly dense. DBSCAN considers two parameters such as:

Eps: The maximum value of radius from its neighborhood.

MinPts: The Eps is surrounded by data points (i.e. Eps-Neighborhood) that should be minimum.

To define Eps-Neighborhood it should satisfy the following condition,

$$NEps(q) : \{ p \text{ belongs to } D | (p, q) \leq E_{ps} \}.$$

In order to understand the Density Based Clustering let us follow few definitions:

- **Core point:** It is point which lies within E_{ps} and MinPts which are specified by user. And that point is surrounded by dense neighborhood.
- **Border point:** It is point that lies within the neighborhood of core point and

multiple core points can share same border point and this point does not contains dense neighborhood.

- *Noise/Outlier*: It is point that does not belongs to cluster.
- *Direct Density Reachable*: A point p is directly Density Reachable from point q with respect to E_{ps} , $MinPts$ if point p belongs to $NE_{ps(q)}$ and Core point condition i.e. $|NE_{ps(q)}| \geq MinPts$
- *Density Reachable*: A point p is said to Density Reachable from point q with respect to E_{ps} , $MinPts$ if there a chain points such as $p_1, p_2, \dots, p_n$,
 $p_1 = q, p_n = p$ such that p_{i+1} is directly reachable from p_i .

Algorithm

1. In order to form clusters, initially consider a random point say point p
2. The second step is to find the all points that are density reachable from point p with respect to E_{ps} and $MinPts$. The following condition is checked in order to form the cluster
 - a. If point p is found to be core point, then cluster is obtained.
 - b. If point p is found to be border point, then no points are density reachable from point p and hence visit the next point of database.
3. Continue this process until all the points is processed.

Advantages

- It can identify Outlier.
- It does not require number of clusters to be specified in advance.

Disadvantages

- If the density of data keeps changing then efficiency of finding clusters is difficult.
- It does not suit for high quality of data and the user has to specify the parameter in advance.

11.7 LIMITATIONS WITH CLUSTER ANALYSIS

There are two major drawbacks that influence the feasibility of cluster analysis in real world applications in data mining.

- The existing automated clustering algorithms on dealing with arbitrarily shaped data distribution of the datasets.
- The second issue is that, the evaluation of the quality of clustering results by statistics-based methods is time consuming when the database is large, primarily due to the drawback of very high computational cost of statistics-based methods for assessing the consistency of cluster structure between the sampling subsets. The implementation of statistics-based cluster validation methods does not scale well in very large datasets. On the other hand, arbitrarily shaped clusters also make the traditional statistical cluster validity indices ineffective, which leave it difficult to determine the optimal cluster structure.

In addition, the inefficiency of clustering algorithms on handling arbitrarily shaped clusters in extremely large datasets directly impacts the effect of cluster validation, because cluster validation is based on the analysis of clustering results produced by clustering algorithms. Moreover, most of the existing clustering algorithms tend to deal with the entire clustering process automatically, i.e., once the user sets the parameters of algorithms, the clustering result is produced with no interruption, which excludes the user until the end. As a result, it is very hard to incorporate user domain knowledge into the clustering process. Cluster analysis is a multiple runs iterative process, without any user domain knowledge, it would be inefficient and unintuitive to satisfy specific requirements of application tasks in clustering.

11.8 OUTLIER ANALYSIS

In statistics, an outlier is a data point that deviates significantly from the norm. Measuring and execution errors are two possible causes. Outlier analysis, also known as outlier mining, is the process of examining data that stands out for whatever reason. It's impossible to do data analysis without encountering outliers. An outlier is a data point that deviates significantly from the norm. As a result, an outlier is critical since it indicates an experimentation flaw. There are numerous applications for outliers, such as identifying fraud and creating new market trends. Outliers are frequently mistaken for random fluctuations in the data. Outliers, on the other hand, are distinct from random noise in the following way:

- An outlier is a point of observation that differs from the rest because of its location.
- To better detect outliers, noise should be eliminated.

11.8.1 Outliers in Data Mining

For the most part, data mining algorithms ignore outliers like noise and exceptions. However, in certain applications like fraud detection, uncommon occurrences can be just as fascinating as the more common ones, therefore performing an outlier analysis becomes critical.

Outliers in Data Mining can have a variety of causes. Following are a few of these contributing factors:

- Identifying financial fraud such as credit card hacking or other similar scams makes use of this technology.
- It's utilized to keep track of a customer's changing purchase habits.
- It's used to find and report human-made mistakes in typing.
- It's utilized for troubleshooting and identifying problems with machines and systems.

11.8.2 Handling of Outliers in Data Mining

Data Mining must deal with outliers for a variety of reasons. These are a few of the explanations:

- The outcomes of databases are impacted by outliers.
- Outliers frequently produce good or useful discoveries and conclusions, allowing researchers to identify various patterns or trends.

- Even in the world of study, outliers can be useful. They can be a lifesaver when doing research.
- Data mining's most important subfield is outlier analysis.

11.8.3 Outlier Detection

Outliers are generally defined as models that are exceptionally far from the mainstream of data. There is no strict mathematical definition of what alienation is; determining whether an observation is an abstraction is ultimately a subjective exercise. An outlier can be interpreted as data or observation that deviates greatly from the mean of a given protocol or set of data. An exception may occur by accident, but it may indicate a measurement error or the given set of data may have a heavier distribution.

Therefore, outlier detection can be defined as the process of detecting and then excluding outsiders from a given set of data. There are no standardized outlier identification methods because these are mostly dataset-dependent. Outlier detection as a branch of data processing has many applications in data stream analysis.

11.8.4 Outlier Detection Techniques

To identify the exterior in the database, it is important to keep in mind the context and find the answer to the most basic and relevant question: “Why should I find outliers?” The context will explain the meaning of your findings.

Remember two important questions about your database during Outlier Identification:

- (i) What and how many features do I consider for outlier detection? (Similarity/diversity)
- (ii) Can I take the distribution (s) of values for the features I have selected? (Parameter / non-parameter)

There are four Outlier Detection techniques in general.

11.8.4.1 Numeric Outlier

A numerical outlier is a simple, non-standard outlier detection technique in a one-dimensional feature space. Exteriors are calculated by IQR (InterQuartile Range). For example, the first and third quarters (Q1, Q3) are calculated. Outlier is a data point x_i that is out of range.

Using the interquartile amplifier value $k=1.5$, the limits are the typical upper and lower whiskers of a box plot.

This technique can be easily implemented on the KNIME Analytics platform using the Numeric Outliers node.

11.8.4.2 Z-Score

The Z-score technique considers the Gaussian distribution of data. Outliers are data points that are on the tail of the distribution and are therefore far from average.

The z-score of any data point can be calculated by the following expression, after making appropriate changes to the selected feature interval of the dataset:

When calculating the z-score for each sample, a limit must be specified in the data set. Some good ‘thumb rule’ limits may be fixed deviations of 2.5, 3, 3.5, or more.

11.8.4.3 DBSCAN

This outlier detection technique is based on the DBSCAN clustering method. DBSCAN is a non-standard, density-based outlier detection method. Here, all data points are defined as focal points, boundary points, or noise points.

11.8.4.4 Isolated forest

This non-parameter system is suitable for large datasets in one or more dimensional features. Isolation number is very important in this outlier detection technique. Isolation number is the number of divisions required to isolate a data point.

11.8.5 Models for Outlier Detection Analysis

There are many approaches to detecting abnormalities. Outlier detection models can be classified into the following groups:

11.8.5.1 Intensive Value Analysis

Extreme value analysis is the most basic form of outlier detection and is suitable for 1-dimensional data. In this external analysis approach, the largest or smallest values are considered externally. The Z-Test and the Students' T-Test are excellent examples.

These are good heuristics for the initial analysis of data but they are not of much value in multifaceted systems. Extreme value analysis is often used as a final step in interpreting the outputs of other outlier detection methods.

11.8.5.2 Linear Models

In this approach, data is structured outside the lower dimensional substructure using linear interactions. The distance of each data point is calculated for a plane that corresponds to the sub-interval. This distance is used to detect outliers. PCA (primary component analysis) is an example of a linear model for anomaly detection.

11.8.5.3 Probabilistic and Statistical Models

In this approach, probability and statistical models consider specific distributions of data. Expectation-enhancement (EM) methods are used to estimate the parameters of the sample. Finally, they calculate the probability of the member of each data point for the calculated distribution. Points with the lowest probability of membership are marked externally.

11.8.5.4 Proximity-based Models

In this mode, the outliers are designed as points of isolation from other observations. Cluster analysis, density-based analysis, and neighborhood environment are key approaches of this type.

11.8.5.5 Information-theoretical models

In this mode, outliers increase the minimum code length to describe a data set.

11.8.6 Uses for Detecting Outliers in Data Mining

In Data Mining, it is common to utilize Outlier Detection to find anomalies. In data mining, it's utilized to find patterns or trends. The following are some examples of where outlier detection is put to use in data mining:

- Fraud Detection

- Telecom Fraud Detection
- Intrusion Detection in Cyber Security
- Medical Analysis
- Environment Monitoring such as Cyclone, Tsunami, Floods, Drought and so on
- Noticing unforeseen entries in Databases

Check Your Progress 1:

1. Describe the uses of cluster analysis in data mining.
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2. Differentiate between Various Clustering Methods along with their description, advantages, disadvantages and algorithms available.
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.....
3. Briefly discuss Outlier and Outlier Detection.
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11.9 SUMMARY

In this unit, we had studied the introductory topics of clustering, clustering methods, algorithms associated with clustering and outlier analysis.

A group of comparable data objects is classed as a cluster in clustering. A collection of data is referred to as a group. The cluster analysis divides data into groupings based on how closely they resemble one another. A label is assigned to a group once the data has been classified into several groups. By doing the classification, it aids in adaptation to shifts.

Cluster Analysis refers to the process of identifying groupings of items that are similar in some respects but differ from one another in other respects. It is a grouping of comparable items that belongs to the same category. A set of items in which the distance between any two things in the cluster is less than the distance between any two objects in the cluster and any object not placed inside the cluster. High-density region related to other regions in multidimensional space. Cluster analysis has wide applications in data mining, information retrieval, biology, medicine, marketing, and image segmentation.

With the help of clustering algorithms, a user is able to understand natural clusters or structures underlying a data set. For example, clustering can help marketers discover distinct groups and characterize customer groups based on purchasing patterns in business. In biology, it can be used to derive plant and animal taxonomies, categorize genes with similar functionality, and gain insight into structures inherent in populations.

Outlier detection from a collection of datasets is a well-known data mining process. Outliers help in detection of unusual patterns and behaviors of different data points which can give a useful result for the research.

More on Clustering can be studied in your third semester MCS-224 Artificial Intelligence and Machine Learning course.

In this next unit, we will discuss Text and Web Mining.

11.10 SOLUTIONS/ANSWERS

Check Your Progress 1:

1.

Data clustering analysis has a wide range of applications, including image processing, data analysis, pattern identification, and market research. With the use of Data clustering, firms can find new client groupings in their database. Buying patterns can also be used to classify data.

Data mining clustering aids in the classification of animals and plants by utilizing comparable functions or genes in biology. Finding out more about how species are structured becomes easier with this information. Data mining clustering identifies geographic areas. There are regions in the earth observation database that are similar to each other.

In the city, a particular type of dwelling is included in a specific neighborhoods. Information can be discovered more easily by classifying online files using clustering in data mining. Additionally, it's employed in detection software. If a credit card is being used fraudulently, the pattern of deceit can be identified using clustering in data mining.

2.

Clustering Method	Description	Advantages	Disadvantages	Algorithms
Hierarchical Clustering	Based on top-to-bottom hierarchy of the data points to create clusters.	Easy to implement, the number of clusters need not be specified apriori, dendrograms are easy to interpret.	Cluster assignment is strict and cannot be undone, high time complexity, cannot work for a larger dataset	DIANA, AGNES, hclust etc.

Clustering Method	Description	Advantages	Disadvantages	Algorithms
Partitioning methods	Based on centroids and data points are assigned into a cluster based on its proximity to the cluster centroid	Easy to implement, faster processing, can work on larger data, easy to interpret the outputs	We need to specify the number of centroids apriori, clusters that get created are of inconsistent sizes and densities, Effected by noise and outliers	k-means, k-medians, k-modes
Distribution-based Clustering	Based on the probability distribution of the data, clusters are derived from various metrics like mean, variance etc.	Number of clusters need not be specified apriori, works on real-time data, metrics are easy to understand and tune	Complex algorithm and slow, cannot be scaled to larger data	Gaussian Mixed Models, DBCLASD
Density-based Clustering (Model-based methods)	Based on density of the data points, also known as model based clustering	Can handle noise and outliers, need not specify number of clusters in the start, clusters that are created are highly homogenous, no restrictions on cluster shapes.	Complex algorithm and slow, cannot be scaled to larger data	DENCAST, DBSCAN
Constraint Based (Supervised Clustering)	Clustering is directed and controlled by user constraints	Creates a perfect decision boundary, can automatically determine the outcome classes based on constraints, future data can be classified based on the training boundaries	Overfitting, high level of misclassification errors, cannot be trained on larger datasets	Decision Trees, Random Forest, Gradient Boosting

3.

An Outlier is a data object that significantly deviates from normal objects as if it were generated by different mechanism. Outlier is different from noise, since noise is a random error or measured variance and it should be removed before outlier detection. Outlier detection aims to find patterns in data that do not conform to expected behavior.

Outlier detection is one of the important aspects of data mining to find out those objects that differ from the behavior of other objects. Finding outliers from a collection of patterns is a popular problem in the field of data mining. A key challenge with outlier detection is that it is not a well expressed problem like clustering so Outlier Detection as a branch of data mining requires more attention. Outlier Detection methods can identify errors and remove their contaminating effect on the data set and as such to purify the data for processing. Outlier detection is extensively used in a wide variety of applications such as military surveillance for enemy activities to prevent attacks, intrusion detection in cyber security, fraud detection for credit cards, insurance or health care and fault detection in safety critical systems and in various kind of images. It is important in analyzing the data due to the fact that they can translate into actionable information in a wide variety of applications. An irregular traffic pattern occurrence in a computer network could mean that a hacked computer is sending out sensitive data to an unauthorized destination.

11.11 FURTHER READINGS

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